

The Universal Vacuum of the Fourth-Order Scalar Field: Metric Orbits, Fock Sectors, and the Krein Boundary

Abstract

For the free fourth-order scalar field $(\square + m_1^2)(\square + m_2^2)\phi = 0$ we separate three structures that are commonly conflated: the geometry of positive metric operators, the geometry of vacuum states, and the geometry of the dynamics. Modewise, the Bender–Mannheim generator is a pure beam splitter in canonical normal-form coordinates — it carries divergent Cartan distortion while creating no quanta there — and no symplectic frame tames its singular values: particle content is a vacuum-ray functional, not a Cartan-norm functional. The entire family of admissible positive metrics fixes the physical vacuum ray, so the vacuum state is metric-independent; it carries exactly $1/3$ of a quantum per ultraviolet mode pair relative to the two-field Fock vacuum, and its GNS representation is disjoint from the latter in every spatial dimension — an obstruction no choice of metric can cure. All nondegenerate selected vacua lie in one universal ultraviolet-dressed Fock sector for $d \leq 3$, anchored by the $\square^2$ vacuum, with a terminating sector hierarchy labeled by $m_1^2 + m_2^2$ for $4 \leq d < 8$ and by the full mass pair for $d \geq 8$. The selected vacuum’s Wightman function is exactly the spectral two-point function of the fourth-order operator for every mass splitting, and its equal-mass limit coincides with the Krein-space vacuum of Bateman and Turok: metric selection and the spectral condition determine the same quasifree vacuum functional (after matching the action sign and infrared extension), the two quantizations differing only in the completion — positive pseudo-Hermitian for split masses, Krein/Jordan at resonance. Finally the selected state is Hadamard in the natural fourth-order sense: its wavefront set is the standard positive-frequency Hadamard set, with a logarithmic short-distance singularity of universal coefficient $1/(16\pi^2)$ for $\log \rho^2$, and its restriction to the local partial-fraction fields is $(\pm)$ Klein–Gordon–Hadamard. All identities are machine-verified.

1 Introduction

The Pais–Uhlenbeck (PU) oscillator and its field-theoretic parent, the free fourth-order scalar

$$(\square + m_1^2)(\square + m_2^2)\phi = 0, \quad m_1 > m_2 \geq 0, \quad (1)$$

admit two apparently different resolutions of the Ostrogradsky ghost problem. In the PT-symmetric approach of Bender and Mannheim [1, 2], each mode carries a positive metric operator $\eta = e^{-Q}$ that renders the dynamics unitary on a Hilbert space; in the Krein-space approach, recently developed for the pure fourth-order theory $\square^2\phi = 0$ by Bateman and Turok [3], the state space is indefinite and probabilities are controlled by a hidden ghost-parity symmetry. Two companion papers [4, 5] analyzed the modewise PT construction as finite-dimensional symplectic geometry: the positive metric is unique up to a stabilizer orbit,

is selected variationally by minimum Cartan distortion, and the modewise theory is exactly solvable in all the respects used below.

This paper carries that analysis to the field theory, where three different geometries are in play and must not be conflated:

$$\text{metric geometry} \neq \text{vacuum geometry} \neq \text{dynamical geometry}. \quad (2)$$

The metric geometry is the symmetric-space geometry of positive operators, G/K for $G = \text{Sp}(2n, \mathbb{C})$, where the Cartan projection μ measures nonunitarity [5]. The vacuum geometry is the geometry of Gaussian vacuum rays, which is governed by a different quotient: the stabilizer $P_\Omega = \{g : \hat{g}\Omega \in \mathbb{C}^\times \Omega\}$ of the vacuum line is a parabolic-type subgroup, and vacuum rays live in G/P_Ω . There is no map $G/K \rightarrow G/P_\Omega$ (the compact $K \cong \text{Sp}(n)$ does not preserve the vacuum ray; only $K \cap P_\Omega \cong U(n)$ does); the correct geometry is the correspondence $G/K \leftarrow G \rightarrow G/P_\Omega$ of two inequivalent quotient maps. The dynamical geometry is the spectral structure of the Hamiltonian, which degenerates (Jordan blocks) exactly at the resonant point $m_1 = m_2$ [4]. Our results, in brief:

1. *Cartan–parabolic decoupling* (Section 2). In canonical normal-form coordinates the Bender–Mannheim generator is a pure beam splitter, $Q' = r(a_1 a_2^\dagger + a_1^\dagger a_2)$: it lies in the complexified number-preserving (Levi) algebra of P_Ω , fixes the canonical vacuum exactly, and yet has Cartan projection (r, r) with $r = \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2} \rightarrow \infty$ in the ultraviolet. Moreover no symplectic frame tames the singular values. Hence there is no inequality of the form vacuum excitation $\geq f(\|\mu\|)$: Cartan distortion does not control particle content.
2. *Orbit constancy and a universal obstruction* (Section 3). The physical annihilator covectors are eigenvectors of the stabilizer generators, so the entire admissible metric family — the Gaussian orbit and all $W = f(N_1, N_2) > 0$ — fixes the physical vacuum ray. The vacuum state is therefore metric-independent. Its occupation relative to the two-field Fock vacuum is exactly

$$\langle N \rangle(k) = \frac{\omega_2(\omega_1^2 + \omega_2^2)}{2\omega_1(\omega_1^2 + \omega_1\omega_2 + \omega_2^2)} \xrightarrow{k \rightarrow \infty} \frac{1}{3}, \quad (3)$$

a universal constant; the total relative quanta scale as $\Theta(V\Lambda^d)$ and the representations are disjoint in every spatial dimension. No choice of metric can cure this.

3. *One universal dressed sector; terminating hierarchy* (Section 4). Selected vacua of different mass pairs satisfy, in common field variables,

$$1 - f(k) = \frac{(\Sigma - \bar{\Sigma})^2}{12k^4} + O(k^{-6}), \quad \Sigma = m_1^2 + m_2^2, \quad (4)$$

with no k^{-6} term and next obstruction $\frac{29}{576}(\delta^2 - \bar{\delta}^2)^2 k^{-8}$, $\delta = \frac{1}{2}(m_1^2 - m_2^2)$. Hence: one universal sector for $d \leq 3$, anchored by the $\square^2$ vacuum; sectors labeled by Σ for $4 \leq d < 8$; by the full mass pair for $d \geq 8$; and the classification terminates.

4. *The spectral bridge* (Section 5). The selected vacuum's Wightman function is exactly the spectral two-point function of (1),

$$\widetilde{W}_{12}^+(p) = \theta(p^0) \frac{\delta(p^2 - m_1^2) - \delta(p^2 - m_2^2)}{m_1^2 - m_2^2}, \quad \widetilde{W}_{mm}^+(p) = -\theta(p^0) \delta'(p^2 - m^2), \quad (5)$$

for all $\Delta = m_1^2 - m_2^2 \geq 0$; the confluent case is the Krein vacuum of [3]. Metric selection and the spectral condition determine the same quasifree vacuum functional; the quantizations differ only in the completion — positive η -inner product where it exists ($\Delta > 0$), Krein space with ghost parity where it must ($\Delta = 0$). The state is analytic across the resonant boundary while the metric and diagonalizability fail there.

5. *Fourth-order Hadamard property* (Section 6). W_{12}^+ is a bisolution with the correct fourth-order commutator, its wavefront set is the standard positive-frequency Hadamard set $\mathcal{C}^+$, and its short-distance singularity is logarithmic with universal coefficient: the $1/\rho^2$ singularity of the Klein–Gordon factors is mass-independent and cancels, leaving

$$W_{12}^+(x) = \frac{1}{8\pi^2} \log \rho + (\text{smooth}), \quad (6)$$

i.e. coefficient $1/(16\pi^2)$ for $\log \rho^2$. Its restrictions to the local partial-fraction fields are exactly $(\pm)$ Klein–Gordon Hadamard functions. Wick powers and perturbation theory can be based on the fourth-order parametrix $H_P = (H_{m_1} - H_{m_2})/(m_1^2 - m_2^2)$.

Together these give the corrected physical picture: *the universal selected vacuum is a fourth-order Hadamard quasifree functional; its apparently infinite particle content relative to the split two-field vacuum reflects inequivalent complex structures, not pathological local ultraviolet singularities* — and the Krein formulation is not the remnant of a divergent positive theory but a different completion of the same, analytically continued vacuum.

Verification. Every identity below — the beam-splitter identity, the occupation (3), the eigenvector/constancy lemma, the sector expansions with their exact coefficients, the Wightman functions and their confluent limits, the commutator normalizations, and the logarithmic coefficient — is machine-verified symbolically, with independent numerical checks; the scripts accompany the repository of [4, 5].

Conventions. Modewise notation follows [4, 5]: for spatial momentum k , the mode is a PU pair with $\omega_i(k) = \sqrt{k^2 + m_i^2}$, canonical variables $V = (x, y, p, q)$ related to the field variable by $z = iy$ (the PT rotation), normalized coordinates with $d_x d_y = \omega_1 \omega_2$, rapidity $r(k) = \log \frac{\omega_1 + \omega_2}{\omega_1 - \omega_2} = \log \frac{(\omega_1 + \omega_2)^2}{\Delta}$, selected diagonalizer $S_+ = e^{(r/2)B}$ and metric $\eta = e^{-Q}$, $Q = \alpha p q + \beta x y$. We write $\Delta = m_1^2 - m_2^2$, $\Sigma = m_1^2 + m_2^2$, $\Pi = m_1^2 m_2^2$.

2 Metric geometry versus vacuum geometry

2.1 The beam-splitter identity

In the canonically normalized coordinates the mode annihilation operators of the auxiliary canonical vacuum are a'_j with $x' = (a'_1 + a_1^\dagger)/\sqrt{2}$, etc. A one-line computation (different modes commute, so no ordering terms arise) gives the exact operator identity

$$Q' = r(x'y' + p'q') = r(a'_1 a_2'^\dagger + a_1'^\dagger a'_2). \quad (7)$$

Thus Q' is a pure beam splitter: $[Q', N_{\text{tot}}] = 0$ and $Q' \Omega_{\text{can}} = 0$, so

$$e^{-Q'/2} \Omega_{\text{can}} = \Omega_{\text{can}} \quad \text{exactly,} \quad (8)$$

even though the Cartan projection of the implemented symplectic map is $\mu(S_+) = (r, r)$ with $r(k) \sim 2 \log k \rightarrow \infty$. Equivalently: Q' lies in the complexified number-preserving algebra, the Levi component of the parabolic-type stabilizer P_Ω of the vacuum line, on which the Cartan norm is unbounded.

2.2 No frame tames the map; the occupation is a state functional

One might hope that a frequency-adapted symplectic frame T_ω (the one whose canonical vacuum is the physical two-field vacuum) turns the divergent Cartan data into bounded physical squeezing. It cannot:

Lemma 2.1 (No frame tames the map). *For every invertible frame T ,*

$$\|\mu(T^{-1}S_+T)\|_\infty \geq \log \varrho((T^{-1}S_+T)^2) = \log \varrho(S_+^2) = r, \quad (9)$$

where ϱ is the spectral radius: the largest singular value of any matrix dominates its spectral radius, and similarity preserves the spectrum of the positive S_+ , which is $\{e^{\pm r/2}\}$. Hence $\|\mu(T_\omega^{-1}S_+T_\omega)\| \rightarrow \infty$ in the ultraviolet for every choice of frames T_ω . (Numerically the physical-frame Cartan data are in fact close to (r, r) ; the lemma is what the theorems below use.)

What is bounded is the *state*: the selected vacuum, computed exactly in [5] as the Gaussian ground state of the PT mode Hamiltonian, has occupation (3) relative to the two-field vacuum, saturating at $1/3$. The two statements are compatible because particle content depends on the transformation only through the vacuum ray it produces — a G/P_Ω datum — while μ is a G/K datum, and the two are inequivalent quotients of G .

Theorem 2.2 (Cartan–parabolic decoupling). *Let $G = \mathrm{Sp}(2n, \mathbb{C})$ act metaplectically on Fock space, and let $q_K : G \rightarrow G/K$ and $q_\Omega : G \rightarrow G/P_\Omega$, $q_\Omega(g) = [\hat{g}\Omega]$, be the two quotient maps. Then: (i) positive-metric geometry is a function of q_K (the Cartan projection); (ii) Gaussian vacuum rays are a function of q_Ω ; (iii) there is no map $G/K \rightarrow G/P_\Omega$ intertwining them, since $K \not\subseteq P_\Omega$ ($K \cap P_\Omega \cong U(n) \subsetneq K \cong \mathrm{Sp}(n)$); (iv) the Cartan norm is unbounded on P_Ω (already on its Levi component), so fibres of q_Ω contain elements of arbitrarily large Cartan norm; consequently (v) no bound of the form $\langle N \rangle \geq f(\|\mu\|)$ or $\leq f(\|\mu\|)$ holds in general. The PU metric orbit realizes the extreme case: $q_\Omega(S_+H) = \{[\Omega_{\mathrm{PT}}]\}$ is a single vacuum ray (Theorem 3.2), while $q_K(S_+H)$ is unbounded, with $\mu(S_+) = (r, r)$ divergent in the ultraviolet yet identically vanishing canonical-frame excitation, and bounded physical excitation (3) despite Lemma 2.1.*

The proof of (iv) is the identity (7) together with $\mu(e^{(r/2)B}) = (r, r)$; (v) follows from the exhibited extremes. This corrects a natural but false expectation — used, e.g., to estimate Bogoliubov costs by $\sinh^2$ of Cartan coordinates — and it is the conceptual reason the theorems below must be formulated at the level of vacuum functionals.

3 Orbit constancy and the universal Fock obstruction

Lemma 3.1 (Eigenvector lemma). *Let ℓ_j be the physical annihilator covectors (the two-field vacuum modes) and X_j the stabilizer generators of [5]. Then*

$$\ell_j^\top X_j = -i \ell_j^\top, \quad \text{hence} \quad \ell_j^\top C(\theta_1, \theta_2)^{-1} = e^{i\theta_j} \ell_j^\top \quad (\theta_j \in \mathbb{C}). \quad (10)$$

Theorem 3.2 (Metric-independence of the vacuum). *The selected physical vacuum state is the same for every admissible positive metric: (i) on the Gaussian orbit η_C , $C \in SO(2, \mathbb{C})^2$, the annihilation conditions defining the vacuum are unchanged by Lemma 3.1; (ii) the metaplectic implementers of the stabilizer are of the form $\exp[\theta_1(N_1 + \frac{1}{2}) + \theta_2(N_2 + \frac{1}{2})]$ and act on the vacuum ray by scalars; (iii) for the full metric family $\eta' = \rho^\dagger W \rho$ with $W = f(N_1, N_2) > 0$ [4], $W^{\pm 1/2}$ also fixes the vacuum ray. Hence $[\Omega_\eta] = [\Omega_{\eta_+}]$ for every admissible η .*

The theorem asserts constancy of the vacuum ray; metric changes do alter the physical inner product and may alter representatives of observables, but any quantity determined by the fixed vector relative to the fixed two-field reference — in particular the occupation (3) and the fidelity below — is metric-independent.

Theorem 3.3 (Universal Fock obstruction). *For every admissible positive metric, in a volume V with cutoff Λ : the relative quanta obey $N(\Lambda) \sim C_d V \Lambda^d$ with $C_d > 0$; the vacuum fidelity obeys $|\langle \Omega_{2\text{-field}} | \Omega_{\text{PU}} \rangle|^2 \leq e^{-cV\Lambda^d}$ (per-mode fidelity $\rightarrow \sqrt{3}/2$ [5]); and in the thermodynamic/continuum limit the GNS representations are disjoint, in every spatial dimension $d \geq 1$. Metric ambiguity cannot cure the inequivalence.*

Proof. Combine Theorem 3.2 with the exact modewise data (3) and the von Neumann infinite-tensor-product criterion, as in [5]. $\square$

4 The universal dressed sector

Comparisons between selected vacua of different mass pairs are made in common physical variables ($\gamma = 1$ PU variables per mode; the normalization map contributes only $O(k^{-2})$). The exact modewise Gaussian of [5] gives, for pairs (m_1, m_2) and $(\bar{m}_1, \bar{m}_2)$, the fidelity expansion (4); at matched Σ the expansion continues

$$1 - f(k) \Big|_{\Sigma=\bar{\Sigma}} = \frac{29}{576} \frac{(\delta^2 - \bar{\delta}^2)^2}{k^8} + O(k^{-10}), \quad \delta^2 - \bar{\delta}^2 = \bar{\Pi} - \Pi, \quad (11)$$

with no k^{-6} term.

Theorem 4.1 (Sector classification). *Let d be the spatial dimension. Relative Hilbert–Schmidt convergence $\int^\Lambda k^{d-1}(1 - f(k)) dk < \infty$ holds iff the first nonvanishing invariant difference is compatible with d :*

spatial dimension	UV sector invariant of selected vacua
$d \leq 3$	none (one universal sector)
$4 \leq d < 8$	$\Sigma = m_1^2 + m_2^2$
$d \geq 8$	(Σ, Π) , i.e. the full mass pair

The hierarchy terminates: two symmetric invariants determine the pair. At the critical dimensions the obstruction is logarithmic rather than power-law: at $d = 4$ (respectively $d = 8$) the divergent sum for $\Sigma \neq \bar{\Sigma}$ (respectively $\Pi \neq \bar{\Pi}$ at matched Σ) grows as $\log \Lambda$, so the critical dimensions belong to the labeled regimes as indicated. In $d \leq 3$ the universal sector is anchored by the $\square^2$ selected vacuum ($m_1 = m_2 = 0$; modewise Gaussian $A = (2k, k^2; k^2, 2k^3)$), relative to which $1 - f = \Sigma^2/(12k^4) + \dots$; all massive selected vacua are quasi-equivalent to it and to each other, while remaining collectively disjoint from the naive two-field sector by Theorem 3.3.

At the resonant boundary three kinds of regularity must be distinguished. *Distributional regularity*: the selected two-point functional is analytic in Δ across $\Delta = 0$ (modewise, $1-f \sim \varepsilon^4$ for $m_{1,2} = m \pm \varepsilon$; the Wightman functions of Section 5 converge smoothly), and the occupation limits commute ($\langle N \rangle \equiv \frac{1}{3}$ at $\Delta = 0$ for every k). *Representation regularity*: for $d \leq 3$ the resonant vacuum lies in the same dressed Fock sector as every nondegenerate selected vacuum (Theorem 4.1). *Dynamical and metric singularity*: as $\Delta \rightarrow 0$ the metric operator degenerates ($\alpha, \beta \rightarrow \infty$), the similarity transformation diverges ($S_+ \sim \Delta^{-1/2}$), and the dynamics becomes nondiagonalizable (Jordan blocks; no positive invariant form [4]). In summary,

$$\boxed{\text{resonance is not a vacuum singularity; it is a metric and dynamical singularity.}} \quad (12)$$

5 The spectral bridge to the Krein quantization

The physical Wightman function of the selected vacuum follows from the exactly solvable mode: with $\tilde{V} = S_+ V$ evolving under the diagonalized Hamiltonian, $W(t) = S_+ e^{tA_0} \Gamma S_+^\top$ pulled back to the field variable $z = iy$ gives, exactly,

$$W_{zz}(t) = \frac{1}{\Delta} \left[\frac{e^{-i\omega_1 t}}{2\omega_1} - \frac{e^{-i\omega_2 t}}{2\omega_2} \right], \quad W_{zz}(t) \Big|_{\Delta \rightarrow 0} = -\frac{(1+i\omega t)e^{-i\omega t}}{4\omega^3}. \quad (13)$$

Both expressions contain only positive frequencies; the first is the momentum-space spectral function (5), the second its confluent (δ') limit. The commutator parts reproduce the classical fourth-order Green-function difference $E_P = [\sin \omega_2 t / \omega_2 - \sin \omega_1 t / \omega_1] / \Delta$ with the normalization $E_P(0) = E'_P(0) = E''_P(0) = 0$, $E'''_P(0) = 1$, and confluent limit $(\sin \omega t - \omega t \cos \omega t) / (2\omega^3)$.

Theorem 5.1 (Bridge theorem). *The selected PU quantization and the Bateman–Turok spectral quantization determine the same quasifree two-point functional for all $\Delta \geq 0$, after matching the overall action sign and (in the doubly massless case) the infrared extension: the selected vacuum’s two-point distribution is (5), whose $\Delta = 0$ limit is the Bateman–Turok $\tilde{W}(p) = \theta(p^0) \delta'_1(p^2)$ [3], realized there by the Jordan mode doublet e^{-ipx} , $(1+2i|p|t)e^{-ipx}$ with null pairing $[a_1, a_2^\dagger] = [a_2, a_1^\dagger] \neq 0$. For $\Delta > 0$ this functional is realized through a positive pseudo-Hermitian completion (η -inner product); at $\Delta = 0$ it is realized through the resonant Krein/Jordan completion with ghost parity. We do not assert positivity with respect to a common involution: “vacuum functional” refers to the normalized quasifree functional determined by the two-point distribution on the fixed field algebra.*

Proof. The first statement is the computation (13) (machine-verified, including the commutator normalization which fixes the action-sign convention: the residual overall sign relative to [3] is exactly the sign of their perfect-square action). Quasifree functionals are determined by their two-point distributions, so equality of (13) with the Bateman–Turok mode Wightman function — both equal to $(1+i\omega t)e^{-i\omega t} / (4\omega^3)$ in matched conventions — identifies the functionals. $\square$

Remark 5.2. (i) The theorem upgrades the resonant-boundary picture: the Krein formulation is not the residue of a positive theory destroyed by an ultraviolet particle catastrophe (Section 2 shows no such catastrophe exists); it is a different completion of the *same*, analytically continued vacuum functional, adapted to Jordan dynamics. (ii) At $\Delta > 0$ the completion ambiguity is invisible in expectation values but material for which operators are Hermitian;

at $\Delta = 0$ positivity is unavailable (no positive invariant metric, [4]) and the Krein completion is the only one. (iii) What remains genuinely open at the resonant point is the treatment of the generalized (Jordan) excitations, where the two completions differ by construction; ghost parity governs precisely that sector [3].

6 Hadamard structure of the selected vacuum

Let W_m^+ denote the Minkowski Klein–Gordon positive-frequency two-point function and define the divided difference and its confluent limit

$$W_{12}^+ = \frac{W_{m_1}^+ - W_{m_2}^+}{m_1^2 - m_2^2}, \quad W_{mm}^+ = -\partial_{m^2} W_m^+ \quad (\text{up to the sign convention of (5)}), \quad (14)$$

which by Theorem 5.1 are the selected two-point functions.

Theorem 6.1 (Fourth-order Hadamard property). *Let $P = (\square + m_1^2)(\square + m_2^2)$, $d = 3$ spatial dimensions. Then:*

1. $P_x W_{12}^+ = P_y W_{12}^+ = 0$, and $W_{12}^+(x, y) - W_{12}^+(y, x) = iE_P(x, y)$ with $E_P = (E_{m_1} - E_{m_2})/\Delta$ the fourth-order commutator function;
2. $\text{WF}(W_{12}^+) = \mathcal{C}^+$, the standard positive-frequency Hadamard wavefront set; the multiplicity of the characteristic polynomial changes the order of the light-cone singularity, not its wavefront directions;
3. the short-distance singularity is logarithmic: with $W_m^+(x) = \frac{1}{4\pi^2} \frac{mK_1(m\rho)}{\rho}$, $\rho^2 = -x^2 + i0x^0$, the mass-independent $\frac{1}{4\pi^2\rho^2}$ term cancels in the divided difference and

$$W_{12}^+(x) = \frac{1}{8\pi^2} \log \rho + (\text{smooth}), \quad (15)$$

the coefficient being universal (mass-independent) and nonzero; the confluent case has the identical coefficient;

4. in the doubly massless case $P = \square^2$ the infrared extension of $\theta(p^0)\delta'(p^2)$ requires a scale; different admissible scales change W by a smooth constant, affecting neither the wavefront set nor the local Hadamard property — the bridge of Theorem 5.1 presumes matched extension conventions;
5. any two fourth-order spectral Hadamard functionals with the same commutator differ by a smooth bisolution; the fourth-order parametrix $H_P = (H_{m_1} - H_{m_2})/\Delta$ (confluently $\partial_{m^2} H_m$) has $W_{12}^+ - H_P$ smooth, so Wick powers $:\phi^n:_{H_P}$ and covariant perturbation theory are available.

Proof sketch. (1) is verified exactly modewise (both branches and the confluent Jordan mode), with the Green normalization $E_P'''(0) = 1$. For (2), the momentum-space form (5) is a distribution supported on the mass shells inside the closed forward cone, giving $\text{WF} \subseteq \mathcal{C}^+$ by the standard conic-support argument; alternatively, divided differences and ∂_{m^2} of the smooth family $m^2 \mapsto W_m^+$ preserve the Hadamard wavefront bound. Equality follows because the coefficient in (3) is nonzero and the on-shell δ/δ' content is nonvanishing on the cone. (3) is the

Bessel expansion $mK_1(m\rho)/\rho = \rho^{-2} + \frac{m^2}{2} \log(m\rho/2) + \dots$: the ρ^{-2} terms cancel and the $\log \rho$ coefficient of the divided difference is $[\frac{m_1^2}{2} - \frac{m_2^2}{2}]/\Delta \cdot \frac{1}{4\pi^2} = \frac{1}{8\pi^2}$ (machine-verified, as is the confluent case). (4) is the observation that the scale enters only through $\log(\mu\rho)$, so a change of scale adds $\log(\mu/\bar{\mu})/(8\pi^2)$, a constant. (5) follows from (1)–(3) by the usual difference argument: the difference of two such functionals is a bisolution with vanishing antisymmetric part and empty wavefront set. $\square$

Proposition 6.2 (($\pm$)Hadamard split structure). *Let $\phi_1 = (\square + m_2^2)\phi/(m_2^2 - m_1^2)$ and $\phi_2 = (\square + m_1^2)\phi/(m_1^2 - m_2^2)$ be the local partial-fraction fields. In the selected vacuum, exactly,*

$$W_{\phi_1\phi_1} = +\frac{W_{m_1}^+}{\Delta}, \quad W_{\phi_2\phi_2} = -\frac{W_{m_2}^+}{\Delta}, \quad W_{\phi_1\phi_2} = 0. \quad (16)$$

The ghost branch carries a Klein–Gordon–Hadamard *distribution with negative Krein signature* — not a positive KG state under the ordinary KG $*$ -structure: the Krein signature is intrinsic to the selected functional, while its local singularity structure is standard on both branches. This makes the summary statement precise:

The universal selected vacuum is a fourth-order Hadamard quasifree functional. Its apparent infinite particle content relative to the split two-field vacuum reflects inequivalent complex structures — the positive-positive versus positive-negative pairing of the two branches — not pathological local ultraviolet singularities.

The logarithmic singularity is the expected one for a dimension-zero field in four spacetime dimensions and is milder than the Klein–Gordon $1/\sigma$; local Wick composites exist, and derivative observables (in particular the stress tensor) require the usual local subtractions relative to H_P , with no evident ultraviolet obstruction. A systematic treatment along the lines of [8, 9, 10] for the fourth-order operator is now well-posed.

7 Discussion

What the three geometries settle. The separation (2) resolves several tensions in the literature at once. The divergent rapidity $r(k) \sim 2 \log k$ is real, but it is metric geometry: it makes the phase-space (observable) metric a Sobolev pair $H^1 \oplus H^{-1}$ [5] and the Dyson operator unboundedly non-unitary in every frame — yet it creates no quanta, because it acts inside the vacuum-line stabilizer. The physical inequivalence to the naive two-field Fock space is vacuum geometry: an $O(1)$ -per-mode, metric-independent statement (Theorems 3.2, 3.3). The resonant $\Delta \rightarrow 0$ singularity is dynamical geometry: Jordan blocks and the death of the positive metric, with the vacuum analytic throughout (Theorem 4.1 ff.). Conflating any two of these produces false conclusions — e.g. ultraviolet particle catastrophes (Λ^{d+2} scalings) or mass-splitting superselection sectors, both of which are refuted by the exact computations here.

Relation to Bender–Mannheim and Bateman–Turok. Theorem 5.1 shows the two programs select the same state, from opposite directions: positivity of the inner product modewise (Bender–Mannheim), spectral positivity of the energy covariantly (Bateman–Turok). The difference is the completion, and the difference matters exactly where the completions are forced

apart: at the resonant point, where positivity fails (the Jordan no-go of [4]) and ghost parity takes over the excitation sector. A sharp remaining question is the representation-theoretic status of the generalized Jordan excitations at $\Delta = 0$: the vacuum sector is common, the one-particle completions are not, and the interacting perfect-square theory of [3] lives precisely there.

Toward quadratic gravity. Applied polarization by polarization, the free-field statements transfer to the scalar and spin-two PU branches of quadratic gravity in its conformally flat and linearized regimes: the physical vacuum is spectral, fourth-order Hadamard, metric-independent, and not a state in the naive two-field (graviton plus ghost) Fock space; PT-based unitarity claims should accordingly be phrased in the dressed representation, and interacting constructions must fix the representation before the perturbative expansion. We leave the interacting analysis, the stress-tensor subtractions relative to H_P , and curved backgrounds to future work.

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